

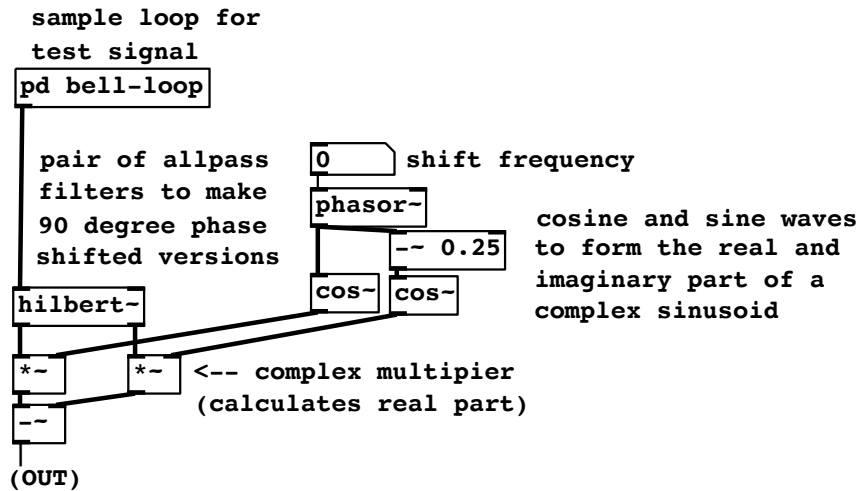


Figure 8.31: Using an all-pass filter network to make a frequency shifter.

all the partials except for the DC one, which we then harvest. This technique is the basis of Fourier analysis, the subject of Chapter 9.

Single sideband modulation

As described in Section 8.4.3, a pair of all-pass filters can be constructed to give roughly $\pi/2$ phase difference for positive frequencies and $-\pi/2$ for negative ones. The design of these pairs is beyond the scope of this discussion (see, for instance, [Reg93]) but Pd does provide an abstraction, `hilbert~`, to do this. Example H09.ssb.modulation.pd, shown in Figure 8.31, demonstrates how to use the `hilbert~` abstraction to do signal sideband modulation. The Hilbert transform dates to the analog era [Str95, pp.129-132].

The two outputs of `hilbert~`, considered as the real and imaginary parts of a complex-valued signal, are multiplied by a complex sinusoid (at right in the figure), and the real part is output. The components of the resulting signal are those of the input shifted by a (positive or negative) frequency specified in the number box.

Using elementary filters directly: shelving and peaking

No finite set of prefabricated filters could fill every possible need, and so Pd provides the elementary filters of Sections 8.2.1-8.2.3 in raw form, so that the user can supply the filter coefficients explicitly. In this section we will describe patches that realize the shelving and peaking filters of Sections 8.3.3 and 8.3.5 directly from elementary filters. First we introduce the six Pd objects that realize elementary filters:

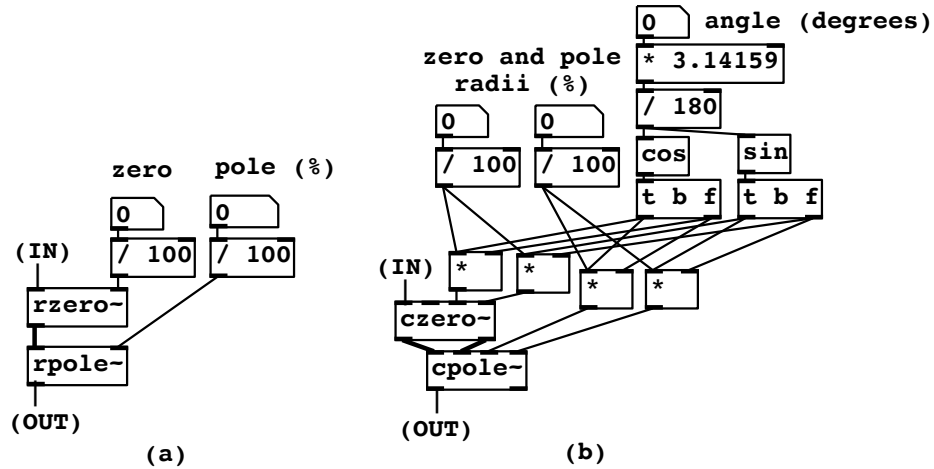


Figure 8.32: Building filters from elementary, raw ones: (a) shelving; (b) peaking.

`rzero~`, `rzero_rev~`, `rpole~`: elementary filters with real-valued coefficients

operating on real-valued signals. The three implement non-recirculating filters of the first and second types, and the recirculating filter. They all have one inlet, at right, to supply the coefficient that sets the location of the zero or pole. The inlet for the coefficient (as well as the left inlet for the signal to filter) take audio signals. No stability check is performed.

`czero~`, `czero_rev~`, `cpole~`: elementary filters with complex-valued

coefficients, operating on complex-valued signals, corresponding to the real-valued ones above. Instead of two inlets and one outlet, each of these filters has four inlets (real and imaginary part of the signal to filter, and real and imaginary part of the coefficient) and two outlets for the complex-valued output.

The example patches use a pair of abstractions to graph the frequency and phase responses of filters as explained in Example H10.measurement.pd. Example H11.shelving.pd (Figure 8.32, part a) shows how to make a shelving filter. One elementary non-recirculating filter (`rzero~`) and one elementary recirculating one (`rpole~`) are put in series. As the analysis of Section 8.3.9 might suggest, the `rzero~` object is placed first.

Example H12.peaking.pd (part (b) of the figure) implements a peaking filter. Here the pole and the zero are rotated by an angle ω to control the center frequency of the filter. The bandwidth and center frequency gain are equal to the shelf frequency and the DC gain of the corresponding shelving filter.

Example H13.butterworth.pd demonstrates a three-pole, three-zero Butterworth shelving filter. The filter itself is an abstraction, `butterworth3~`, for easy

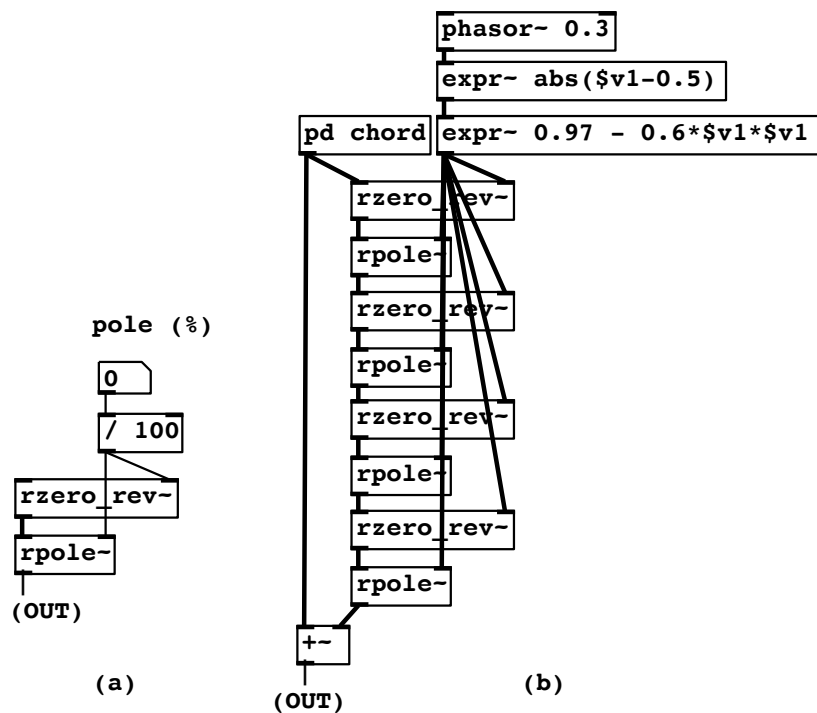


Figure 8.33: All-pass filters: (a) making an all-pass filter from elementary filters; (b) using four all-pass filters to build a phaser.

reuse.

Making and using all-pass filters

Example H14.all.pass.pd (Figure 8.33, part a) shows how to make an all-pass filter out of a non-recirculating filter, second form (`rzero_rev~`) and a recirculating filter (`rpole~`). The coefficient, ranging from -1 to 1, is controlled in hundredths.

Example H15.phaser.pd (part b of the figure) shows how to use four all-pass filters to make a classic phaser. The phaser works by summing the input signal with a phase-altered version of it, making interference effects. The amount of phase change is varied in time by varying the (shared) coefficient of the all-pass filters. The overall effect is somewhat similar to a flanger (time-varying comb filter) but the phaser does not impose a pitch as the comb filter does.

Exercises

1. A recirculating elementary filter has a pole at $i/2$. At what angular frequency is its gain greatest, and what is the gain there? At what angular frequency is the gain least, and what is the gain there?
2. A shelving filter has a pole at 0.9 and a zero at 0.8. What are: the DC gain; the gain at Nyquist; the approximate transition frequency?
3. Suppose a complex recirculating filter has a pole at P . Suppose further that you want to combine its real and imaginary output to make a single, real-valued signal equivalent to a two-pole filter with poles at P and $\bar{P}$. How would you weight the two outputs?
4. Suppose you wish to design a peaking filter with gain 2 at 1000 Hertz and bandwidth 200 Hertz (at a sample rate of 44100 Hertz). Where, approximately, would you put the upper pole and zero?
5. In the same situation, where would you put the (upper) pole and zero to remove a sinusoid at 1000 Hertz entirely, while attenuating only 3 decibels at 1001 Hertz?
6. A one-pole complex filter is excited by an impulse to make a tone at 1000 Hertz, which decays 10 decibels in one second (at a sample rate of 44100 Hertz). Where would you place the pole? What is the value of “q”?